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Multi-channel EEG signal acquisition system based on nRF52832

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***Abstract*—This study examines a portable multi-channel EEG signal acquisition device based on the nRF52832 that can gather human EEG signals. Power supply, signal pre-processing, signal acquisition, the primary controller, and other components make up the majority of the system. The 24-bit analog front-end chip ADS1299 and the Bluetooth control chip nRF52832 make up the majority of the hardware component. The device can receive 16 channels of EEG signal data and communicate that data over Bluetooth to the computer side. The test demonstrates the system's tiny size, low power usage, and ability to lower operating costs.**

Keywords-EEG signals; Bluetooth transmission; ADS1299; Signal Acquisition;

I. INTRODUCTION

The EEG signal is a crucial bioelectrical signal in the human body and contains extensive information regarding brain function. The human EEG signal can react to outside stimuli, provide information about the human body, and also show data about brain disorders [1]. As science and technology have advanced, more people have begun to investigate EEG signals, and their application in the field of brain-computer interface and other areas is spreading rapidly [2].

At present, most EEG signal acquisition devices are large and expensive [3], and most of them have strict requirements on working environment, and the use scenarios of these devices are relatively single. In this study, a portable multi-channel EEG signal acquisition system is developed that can acquire 16 EEG signals at once to satisfy the majority of acquisition point needs. The device is Bluetooth enabled to expand the usage scenarios for EEG signal acquisition. The system has the features of small size, low power consumption, easy application, and rich usage scenarios, which can provide further EEG signal research.

II. OVERALL DESIGN OF THE SYSTEM

A. Generation of EEG signals and their characteristics

The human brain contains a large number of neurons, and when the nerve cells in the brain are active, they generate electrical signals. The energy of the electrical signals generated by a large number of neurons during brain activity will accumulate, and when the accumulated energy exceeds a certain threshold, EEG signals will be formed [4]. The amplitude of EEG signal ranges from 5 to 100 μV, and the main frequency band of the signal is from 0.5 to 100 Hz [5].

EEG signals are characterized by weakness, non-linearity, non-smoothness and randomness, and are easily disturbed by various types of external noise [6]. Meanwhile, EEG signals contain rich frequency components. According to the frequency and amplitude of EEG signals, EEG signals can be classified into five categories, namely δ-wave, β-wave, θ-wave, α-wave and γ-wave [7]. Each category of EEG signals has a unique location of appearance and corresponding mental activity at the moment of appearance. Table 1 displays the classification of EEG waveforms.

TABLE 1. EEG SIGNAL CLASSIFICATION

EEG signal wavefom	**Differentiation**		
	Frequency band (Hz)	***Location of appearance***	***Psychological activities***
Dela(δ)	0.5~3	Occipital lobe, temporal lobe	Slow wave sleep
Theta(θ)	4~7	Occipital lobe, parietal lobe	Unconscious, contemplative sleepiness
Alpha(α)	8~13	Occipital lobe, parieto-occipital lobe	Awake, quiet, eyes closed
Beta(β)	14~30	Frontal, central region	Alertness, thinking, concentration
Gramma(γ)	30~50	----	Excitement

B. EEG signal acquisition electrode conduction method

By studying the ADS1299 data sheet [8], we learned the electrode lead method for EEG signal measurement. This device uses a single-stage lead approach to acquire EEG signals and will employ reference electrodes. The 16 measuring channels' electrode placement follows the 10-20 national standard lead system [9]. A single reference electrode that is shared by all channels will be attached to the body's left ear. When acquiring EEG signals, the ADS1299 uses differential input to measure the input signal. The right leg drive [10] circuit is designed through the bias drive amplifier BIAS [11] built into the ADS1299 chip. The right leg drive circuit inverts the common-mode signal in the human EEG signal to produce a common-mode signal of opposite polarity, which enters the human body through the bias electrode, thereby suppressing the common-mode signal and improving the signal-to-noise ratio. The right leg drive will use a bias electrode that is connected to the right ear of the human body.

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In this study, silver chloride dry electrodes were used to contact the scalp for EEG signal acquisition. Silver chloride electrodes have the advantages of stable potential, good reproducibility, strong electrode structure, and low temperature hysteresis. The electrode is highly resistant to wear and tear and suitable for most of the acquisition environments, plus its stable performance is suitable for EEG signal acquisition.

C. *Overall system structure solution*

EEG signal is an extremely weak physiological electrical signal with low frequency range, easy to be disturbed and non-smooth randomness, etc. We need to amplify the EEG signal before we can observe it. The need to use an integrated chip to capture the EEG signal in its entirety has been made possible by advances in technology.

EEG signal acquisition system includes four parts: EEG signal acquisition, signal reception, signal processing and signal display. The front-end of the signal acquisition part needs to select the appropriate signal pre-processing protection circuit and acquisition circuit to filter, amplify and convert the EEG signal to analog-to-digital. Then the EEG signal data is transmitted to the processor through the hardware transmission interface, and then the data is sent to the PC through Bluetooth communication, and the EEG signal data is received using the upper computer, and then the EEG signal data is processed.

Figure 1 shows the framework of the overall solution structure of this system. The EEG signal acquisition system in this study contains acquisition electrodes, pre-processing circuit, integrated analog front-end ADS1299 acquisition circuit, Bluetooth master control nRF52832, and power management circuit. The whole system. The EEG signal acquisition uses silver chloride electrodes to simultaneously acquire 16 channels of analog EEG signal data, which are analog low-pass filtered by the pre-processing circuit to suppress the interference of high-frequency noise, and then the two integrated analog front-end ADS1299 is used for analog-to-digital conversion to obtain 16 channels of 24-bit digital EEG signal data. The digital signal is then transferred to the nRF52832 Bluetooth master module through the SPI interface, and finally the digital EEG signal data is uploaded to the host computer using Bluetooth communication. The power management circuit uses a lithium battery for power supply and completes charge and discharge management to power each module.

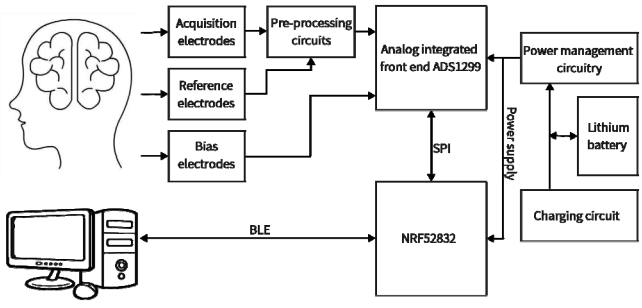


Figure 1. Overall system solution structure

III. HARDWARE DESIGN OF EEG ACQUISITION SYSTEM

A. *Pre-processing part*

The EEG signal is susceptible to interference from other signals, such as human eye, myoelectricity and mains 50Hz interference, due to its weak energy, and the signal-to-noise ratio is very low. The pre-processing circuit is designed in the hardware acquisition system, which completes the pre-processing operation of the simulated EEG signal, filters out the interference signals such as high frequency and industrial frequency, and completes the preliminary EEG signal pre-processing, so as to improve the signal-to-noise ratio of the EEG signal. At the same time, the system also needs to ensure the safety of the system as it acts directly on the human body through EEG electrodes. In this system, low-pass filter is used to filter, low-pass filter consists of capacitor-resistor, which can filter out the high frequency noise interference in the original EEG signal, and the pre-processing protection module is mainly a second-order low-pass filter. The pre-processing protection module is connected to each electrode by chip MMQA5V6T1G to prevent the interference of human static electricity and possible damage caused by electrostatic breakdown during the acquisition process. MMQA5V6T1G is an electrostatic discharge protection chip, which has a unidirectional four-wire ESD protection function and can protect the subsequent acquisition modules.

B. *System power supply part*

The design of the power supply module is crucial for microvoltage signals such as EEG signals. The power supply module can affect the accuracy of the acquisition system. The EEG signal acquisition system in this study uses 3.7V lithium battery power supply and 5V Type-C power supply, the power supply can be switched by a switch, or you can choose to use Type-C to charge the lithium battery, where TP4057 is used for lithium battery charging management. nRF2832 and ADS1299 both use 3.3V digital power supply, while ADS1299 in The LDO chips RT9139_25G, RT9139_33G, TPS72325 and DC-DC chip LM2664 are used to output 3.3V digital power supply and 2.5V and -2.5V analog power supply in bipolar power mode.

C. *Signal acquisition part*

The EEG signal acquisition part of this system uses TI's ADS1299 analog front-end signal acquisition chip [12], which integrates an 8-way low-noise programmable gain amplifier (PGA), an 8-way 24-bit high-precision analog/digital converter ADC, a synchronous sampling frequency of 250 Hz/s to 16 kHz/s and a serial peripheral interface SPI. The ADS1299 chip performs preliminary filtering, signal amplification, and analog-to-digital conversion, and then transmits them to the master chip via the SPI protocol. In this system, two 8-channel ADS1299 analog front-end chips will be used to acquire 16 channels of EEG signals.

There are standard mode and daisy chain mode when multiple ADS1299s work simultaneously. In this system, standard mode is used. The data input pin DIN, data output pin DOUT and clock signal pin SCLK of the SPI interface of the two ADS1299s are the same, and different chip select pins CS

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are used to control the data acquisition transmission of different ADS1299 chips respectively. In the configuration process, in order to make the two ADS1299s complete the work of synchronous acquisition, the two acquisition chips are set to external crystal mode, and the external active crystal is connected to the external clock CLK pins of the two chips at the same time, so as to realize the synchronous acquisition of the two chips, where the clock frequency of the external active crystal is 2.048MHz. The circuit schematic of the acquisition part ADS1299 chip is shown in Figure 2. The pin connections of the two ADS1299 chips are identical except for the CS pin.

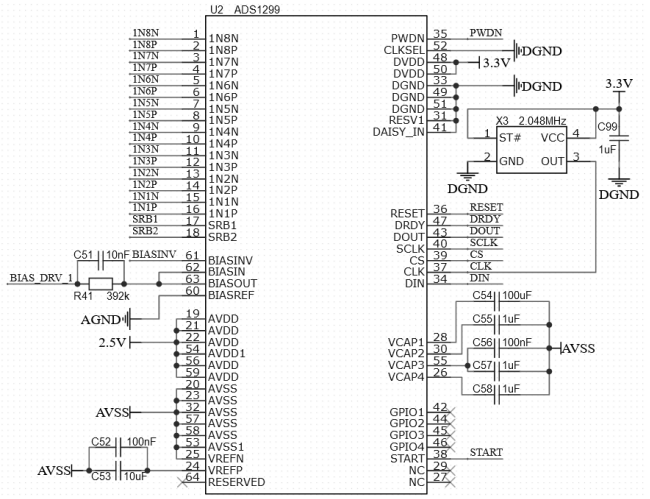


Figure 2. Circuit schematic of ADS1299

D. *Main control part*

The main control part of the system uses Nrodic's nRF52832 low-power Bluetooth chip, which is based on the ARM Cortex-M4F core microprocessor. nRF52832 has ultra-low-power characteristics, supports low-power Bluetooth (BLE), the chip integrates a 2.4 GHz wireless transceiver, memory with 512kB flash and 64kB RAM. nRF52832 peripheral resources are abundant, can be directly connected with ADS1299 through the SPI interface, but also through the SWD serial debugging interface for program download debugging, to ensure the normal operation of the entire system. The chip is equipped with Bluetooth, Bluetooth data transmission can be achieved, which can make the experimental test is no longer limited to the environment, the subject has a greater range of activities, opening up the use of the system scenarios.

IV. EEG ACQUISITION SOFTWARE DESIGN

The development software selected for the lower computer of the acquisition system is Keil uVision5, and the development environment is MDK-ARM. The most important part of the development process is the acquisition part of the ADS1299. The 16-channel data acquisition is achieved by configuring the pins and registers of the two ADS1299s, and the data transfer is performed in strict accordance with the SPI timing. The flow chart of the lower computer of the system is shown in Figure 3.

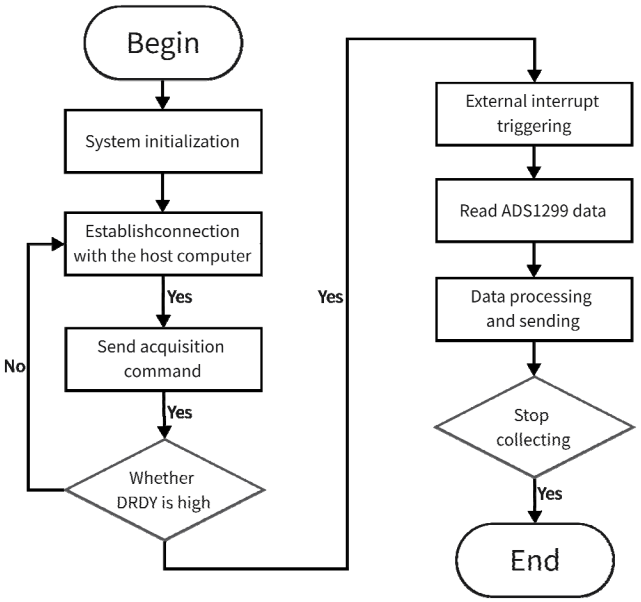


Figure 3. System flow chart for lower computer

The system enters into initialization after power on, which mainly includes SPI initialization, external interrupt initialization, ADS1299 initialization, Bluetooth service initialization to establish connection with the upper computer. ADS1299 initialization configures the internal registers of two ADS1299s separately through SPI for function setting. When the EEG signal data acquisition is completed, the data preparation pin DRDY of the acquisition chip ADS1299 will jump to low level and connect the DRDY pin to the external interrupt pin of the master to detect its falling edge signal, and then read the EEG signal data acquired by ADS1299 through SPI. . The data processing and transmission section converts the raw 24-bit complement data into raw EEG voltage data, and then transmits the 16 channels of raw EEG voltage data to the host computer via Bluetooth serial transmission.

The upper computer of the acquisition system builds the data reception environment by Python software to realize the reception, real-time display and storage of EEG data. MATLAB software is used to build the data processing environment to realize the display and analysis of EEG waveform, and carry out the analysis of EEG signal frequency and other characteristics of any channel. Complete the processing of EEG signal data through MATLAB, including low-pass filtering, industrial frequency trapping and other processing, which makes the EEG signal more pure.

V. EXPERIMENTAL TEST RESULTS

A. *System Implementation*

The finalized multi-channel EEG signal acquisition system is physically shown in Figure 4. The system PCB board is powered by lithium battery, the EEG signal acquisition is realized by multi-channel electrode line, and the Bluetooth transmission of EEG signal data is realized by Bluetooth. The system board is 50 mm long and 30 mm wide, and only needs

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to connect the reference electrode line, bias electrode line and 16 recording electrode lines to be able to acquire 16 channels of EEG signals when used. The system is powered by lithium battery and the board is small and easy to use, no additional connections are needed. The power consumption of ADS1299 does not exceed 40mV, the power consumption of nRF52832 is 17mV, the system can be used for a long time with low power consumption. After testing, the whole system can meet all the requirements of the measurement.

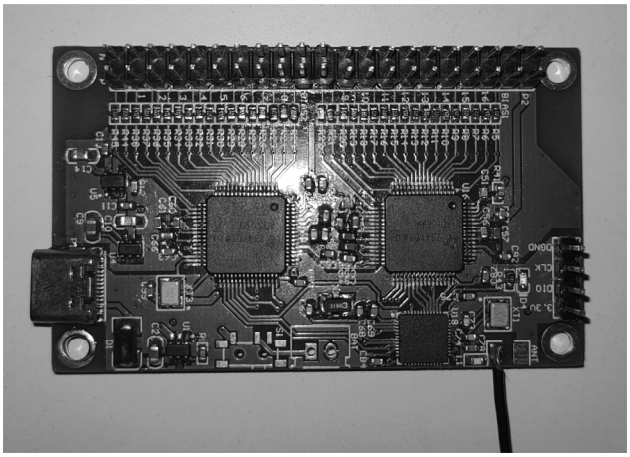


Figure 4. Multi-channel EEG signal acquisition system

B. System acquisition test results

The EEG signal system was tested, and the system boards were tested in accordance with JJG 1043-2008 and JJG 749-2007 standards. The input noise of the system is 1.5uV, the common mode rejection ratio is 95dB, and the input impedance is 110MΩ. The system performance all conforms to the requirements of the standards, and the amplitude and frequency errors of the acquired signals in the accuracy test are negligible, which can meet the acquisition function of EEG signals.Translated with www.DeepL.com/Translator (free version)The 16-channel EEG signal waveforms acquired by the portable EEG acquisition system in a room temperature environment are shown in Figure 5.

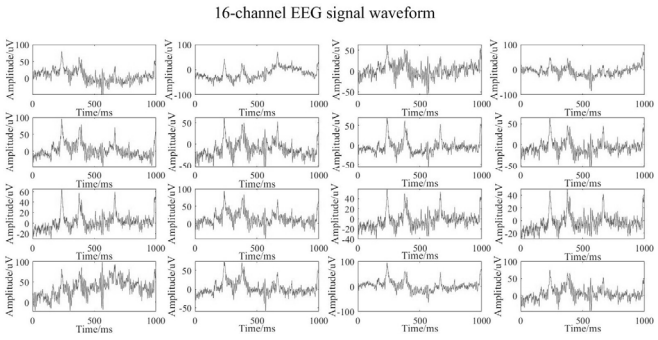


Figure 5. 16-channel EEG signal waveform

VI. SUMMARY AND OUTLOOK

This study designs a portable multi-channel EEG signal acquisition system, using low-power analog front-end chip ADS1299 combined with low-power Bluetooth master chip nRF52832 to complete the acquisition of EEG signals, reducing the overall power consumption of the system and reducing the size of the system, and expanding the use of scenarios by carrying Bluetooth is also convenient for subsequent functional expansion. After testing and verification, the system is able to acquire 16 channels of EEG signals simultaneously with high resolution, which can realize the function of multi-channel EEG signal acquisition. The experimental results show that the system complies with the relevant standards and can provide technical guarantee and reference for the subsequent research.

In addition, the system is slightly inadequate in the acquisition part. The ADS1299 has an internal electrode state detection function, which can be extended to detect lead shedding and improve the system's self-testing error capability. In the future, we will focus on the research of the processing algorithm related to EEG signals to improve the overall use and analysis capability of the system.

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